

YUTONG WU

Irvine, CA | yutonw37@uci.edu | +1 949-981-2959

SUMMARY

I'm a third-year Computer Science student at UC Irvine, interested in exploring how decision making and machine learning can be applied to real-world problems. I have built projects ranging from web-scale search engines to networking systems, and I am now eager to deepen my experience through research in AI and data-driven decision making.

PROFESSIONAL EXPERIENCE

Search Engine Project

- Built a web search engine from scratch capable of handling tens of thousands of web pages under strict operational constraints
- Designed and implemented an inverted index with partial on-disk indexing and multi-pass merging to handle limited memory environments
- Optimized query processing to achieve sub-300ms response time by reading postings directly from disk without loading the full index into memory

Simplified WeChat & WiFi-Roaming Projects

- Developed two Python-based systems: a simplified messaging app and a roaming-aware networking client
- Applied socket programming, GUI design, class protocols, and custom magic methods to build interactive features
- Analyzed and optimized time and space complexity of core algorithms

Planning and Decision-Making (CS171 Project)

- Modeled planning problems using PDDL (Planning Domain Definition Language), specifying objects, states, and actions
- Strengthened understanding of AI reasoning, decision-making under constraints, and classical planning methods

AWARDS

- **Siggraph Global First Place, SpaceTime Competition**
 - **Engineering Inspiration Award, FIRST Robotics Competition**
 - **Global Gold, British Physics Olympiad**
-

SKILLS

Python Programming	Figma	C++ Programming
Data Visualization	SQL	Decision-Making(PDDL)

EDUCATION

University of California, Irvine

B.S. in Computer Science

Expected Graduation: June 2027

current GPA: 3.932/4 Dean's Honor List

- Relevant Coursework: Data Structures, Web Development, Linear Algebra, Algorithms, Statistic, Human-Computer Interaction, Artificial Intelligence, ML, Information Visualization, Search engines
-

ADDITIONAL INFORMATION

- **Languages:** English, Mandarin.
- **Other skills :** Rhino/Blender Modeling, R programming, Procreate